

UBT614: BIOSAFETY, BIOETHICS & IPR

L	T	P	Cr
2	0	0	2.0

Course Objective: To introduce basic concepts of ethics and safety that are essential for all basic and applied sciences. To understand balanced integration of scientific and social knowledge and its protection in sustainable development.

Syllabus

Biosafety: Evolution of the concept of biosafety; biohazard, need and application of biosafety in laboratories and industries; biosafety guidelines, regulations and their implementation; Classification and Description of Biosafety levels; Design of clean rooms and types of biosafety cabinets; Risk assessment and containment levels through case studies; bio-medical and hazardous wastes; Good laboratory practice (GLP); bio-terrorism; Convention on biodiversity and Cartagena protocol on biosafety; types of risk and issues associated with GMO's for animal, human, agriculture and environment.

Bioethics: Introduction and need of bioethics, its relation with other branches, Ethical Issues involving GMOs; ethics related to human cloning, prenatal diagnosis, agriculture and animal rights through case studies; Socio-economic impact of biotechnology.

Intellectual Property Rights (IPR): Introduction to IPR, types of IP (patent, copyrights, geographical indications, trademarks, trade secret, Industrial designs), treaties in IPR, Patent laws, Legislations covering IPR's in India, IPR Protection, patent filing in biotechnology, provisional and complete specification, patentable and non-patentable items.

Laboratory Work (if applicable)

Course Learning Objectives (CLO)

The students will be able to:

1. Interpret basics of biosafety and bioethics and its impact on all the biological sciences and the quality of human life.
2. Recognize importance of biosafety practices and guidelines in research.
3. Comprehend benefits of GM technology and related issues.
4. Recognize importance of protection of new knowledge and innovations and its role in business.

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

Text Books

1. Sateesh, M.K., Bioethics and Biosafety, IK International Publishers (2008).
2. Singh I. and Kaur, B., Patent law and Entrepreneurship, Kalyani Publishers (2006).
3. Srinivasan, K. and Awasthi, H.K., Law of Patents, Jain Book Agency (1997).

Reference Books

1. Narayan, P., Patent Law, Eastern Law House (1975).
2. Jonathan, Y.R., Anthology of Biosafety (Vols. 1-4), American Biological Safety Association (2005).
3. Encyclopedia of Ethical, Legal and Policy issues in Biotechnology, John Wiley & Sons Inc. (2005).

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	MST	25-35
2.	EST	35-45
3.	Sessionals (May include assignments/quizzes)	30

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